



TB AUDIO PLUG-INS
Team Impulse Impact

USER MANUAL

TB Ring Modulation

Ring Modulator

A four-control ring modulator with an internal carrier from 0.1 Hz to 5 kHz, moving from slow tremolo to metallic sidebands.

Carrier 0.1 Hz to 5 kHz

Sine-to-square shape

Wet tone filter

6 factory presets



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01 Overview

TB Ring Modulation multiplies the input by an internal carrier. At low carrier frequencies this sounds like tremolo and pulsing amplitude; at higher frequencies it creates new, often metallic sidebands above and below the original sound.

Four controls cover it: **FREQUENCY** sets the carrier, **SHAPE** morphs its waveform, **TONE** focuses the top end of the modulated signal, and **MIX** places the effect. Parameter and bypass smoothing keep automation clean on mono or stereo sources, and no external routing is needed.

Key features

- Internal carrier from 0.1 Hz to 5 kHz
- SHAPE morphs from sine through triangle to square
- TONE low-pass on the modulated signal
- MIX blends the dry and ring-modulated signals
- Carrier field display with frequency, shape, and waveform
- Smoothed parameters and bypass
- 6 factory presets, user presets, undo, and redo

Signal flow



About this manual

This manual describes TB Ring Modulation 1.0.0 for Windows (VST3). The current macOS build is also version 1.0.0; small differences between platforms are possible.

02 Quick start

- 1 Insert TB Ring Modulation on a vocal, drum, synth, or guitar track. The defaults are **FREQUENCY** 120 Hz, **SHAPE** sine, **TONE** 12 kHz, and **MIX** 50%.
- 2 For tremolo, set **FREQUENCY** **3** below about 20 Hz.
- 3 For metallic or robotic tones, raise **FREQUENCY** into the hundreds or thousands of hertz.
- 4 Make the carrier harsher with **SHAPE** **4**, and tame the top end with **TONE** **5**.

- 5 Blend the result with **MIX** 6.

03 Interface



TB Ring Modulation 1.0 interface, showing the Robot Voice preset.

- 1 **Preset bar** — previous and next preset (hold to browse), the preset list, Save, Undo, and Redo.
- 2 **CARRIER FIELD** — the carrier frequency, its shape, and the waveform.
- 3 **FREQUENCY** — the carrier frequency.
- 4 **SHAPE** — sine, triangle, or square, and the morphs between them.
- 5 **TONE** — low-pass on the modulated signal.
- 6 **MIX** — dry/wet blend.
- 7 **Bypass** — bypasses the complete effect.

04 Controls

Control	Range · default	What it does
FREQUENCY	0.1 Hz to 5 kHz · 120 Hz	The internal carrier frequency. Low values give tremolo and pulsing; high values give inharmonic, metallic sidebands.
SHAPE	0 to 100% · 0% (sine)	Morphs the carrier from sine (0%) through triangle (50%) to square (100%). Toward square there are more harmonics, so check the high end.
TONE	500 Hz to 20 kHz · 12 kHz	Low-pass cutoff on the ring-modulated signal. It does not change the carrier frequency.
MIX	0 to 100% · 50%	Blends the dry and ring-modulated signals. At 100% the original pitch cues change strongly.
Bypass	Off, On · Off	Bypasses the complete effect, with smoothing.

Ring modulation sounds most musical when the carrier is related to the key, for example the root or the fifth. Unrelated frequencies give clangorous, bell-like or robotic results.

05 Presets

Step through presets with the arrows; hold an arrow to browse continuously. Open the list to choose a factory or user preset. Save stores the current settings as a .tbring file. Undo and Redo step through recent changes. The factory presets can also be selected from the host through a program parameter.

Group	Preset	FREQUENCY	SHAPE	TONE	MIX
Start	Init	120 Hz	0% (sine)	12 kHz	50%
Start	Subtle Tremolo	5 Hz	0% (sine)	16 kHz	30%
Metal	Bell Alloy	437 Hz	15%	18 kHz	70%
Metal	Robot Voice	82 Hz	100% (square)	8 kHz	75%
Texture	Radio Sidebands	1250 Hz	50% (triangle)	4.5 kHz	65%
Texture	Digital Insect	2800 Hz	100% (square)	9 kHz	60%

06 Recipes

Starting points for common jobs. Adjust them by ear.

Tremolo

A gentle amplitude pulse on guitar or keys.

Preset Subtle Tremolo

FREQUENCY 2 to 8 Hz

MIX 20 to 40%

- 1 Set FREQUENCY to taste; it is free-running, not tempo-synced.

Robot voice

A buzzing, robotic vocal.

Preset Robot Voice

FREQUENCY 60 to 120 Hz

SHAPE Square

- 1 Lower TONE if the buzz is too harsh.

Bell and metal textures

Clangorous tones on percussion or synths.

Preset Bell Alloy

FREQUENCY 300 to 1000 Hz

- 1 Sweep FREQUENCY slowly to find pleasing sidebands.

Lo-fi radio

A narrow, broken-radio effect.

Preset Radio Sidebands

TONE 3 to 5 kHz

- 1 Combine with a high-pass EQ for a smaller sound.

07 Tips and troubleshooting

Symptom	What to check
The effect is too harsh.	Lower TONE, move SHAPE toward sine, or lower MIX.
The pitch of the source disappears.	That is normal at high MIX and FREQUENCY. Lower MIX to keep the original pitch audible.
Low frequencies sound like a pulse, not a tone.	Below about 20 Hz the carrier works like tremolo. Raise FREQUENCY for sidebands.
Nothing changes.	MIX is at 0% or the effect is bypassed.

08 Parameter reference

Panel label	Host parameter	Range	Default
FREQUENCY	Frequency	0.1 to 5000 Hz	120 Hz
SHAPE	Shape	0 to 100% (sine to square)	0%
TONE	Tone	500 to 20000 Hz	12000 Hz
MIX	Mix	0 to 100%	50%
Bypass	Bypass	Off, On	Off
Preset list	Program	6 factory presets	Init

09 Version history

Version	Changes
1.0.0 · 2026-08-02	First release: an internal carrier from 0.1 Hz to 5 kHz, sine-triangle-square shape morphing, a wet tone filter, and six factory presets.

10 Support and license

Report problems or request features at github.com/TaebeumKim/TB_Audio-Plugins/issues. Include the plug-in version, your DAW, and your operating system.

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